

Generation of SPP on Si targets in water medium upon femtosecond laser irradiation

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The excitation of Surface Plasmon Polariton excitation is investigated for layered media. The resulting periodicity of SPP modes excited in Si into water explain the sub-200 nm LIPSS periodicities obtained experimentally. The interference of SPPs living at different interfaces is investigated and give birth to a wide range of far-subwavelength modes.

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I. THEORY OF SURFACE PLASMON POLARITONS

The SPP excitation conditions are investigated at a simple interface, and then in a system composed of one layer, separating two semi-infinite parts [Figure 1].

A. SPP excitation at a single interface

1. Conditions in a perfect media

Condition for SPP excitation neglecting the absorption of the external medium is given by both conditions [1–3]

$$\begin{cases} f_{real}^{(1)} = \Re(\varepsilon_1) \Re(\varepsilon_2) < 0 \\ f_{real}^{(2)} = \Re(\varepsilon_1) + \Re(\varepsilon_2) < 0 \end{cases} \quad (1)$$

where $\varepsilon_1 \in \mathbb{R}$ and $\varepsilon_2 \in \mathbb{C}$.

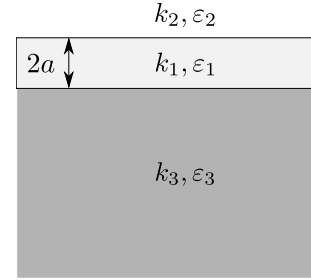


Figure 1: Scheme of the considered multilayer system. $t = 2a$ is the thickness of the layer. k_i are wavenumber in the layer i , and ε_i is the dielectric function used to that layer.

2. Conditions a medium with losses

Condition for SPP excitation at a single interface between media 1 and 2 is given by [4]

$$f(\varepsilon_1, \varepsilon_2) < 0 \quad (2)$$

where

$$f(\varepsilon_1, \varepsilon_2) = \Re(\varepsilon_1) \Re(\varepsilon_2) + \Im(\varepsilon_1) \Im(\varepsilon_2)$$

compatible with recent work of Norrman et al. [5].

3. Periodicity of SPPs at single interface

Periodicity of the SPPs at a single interface is given by $\Lambda = \frac{2\pi}{\Re(\beta)}$ [1–3] where

$$\beta = \frac{\omega}{c} \sqrt{\frac{\varepsilon_1 \varepsilon_2}{\varepsilon_1 + \varepsilon_2}}. \quad (3)$$

This relation is valid at any interface but must be restricted by condition by Eq. (2).

B. Multilayer SPP model

Dispersion relation describing the electric and magnetic fields in the Transverse Magnetic direction, in a multilayer system is given by [3, 6]

$$e^{-2k_1 t} = \frac{\frac{k_1}{\varepsilon_1} + \frac{k_2}{\varepsilon_2}}{\frac{k_1}{\varepsilon_1} - \frac{k_2}{\varepsilon_2}} \times \frac{\frac{k_1}{\varepsilon_1} + \frac{k_3}{\varepsilon_3}}{\frac{k_1}{\varepsilon_1} - \frac{k_3}{\varepsilon_3}} \quad (4)$$

Material	ε_∞	m^*	ν^{-1}	Ref
Si	$13.64 + i0.048$	0.18	1.1 fs	[8]
a-Si	$14.9 + i0.627$	0.20	1.0 fs	[9]
Water	1.7583	1.00	1.7 fs	[10]

Table I: Parameters used in the dielectric functions given by Eq. (6).

where

$$k_i^2 = \beta^2 - k_0^2 \varepsilon_i \quad (5)$$

$k_i, i \in \{1, 2, 3\}$ is the complex wave number in medium i . β is the complex wavenumber characterizing the possible electromagnetic modes in the thin layer. $\varepsilon_i, i \in \{1, 2, 3\}$ are the dielectric functions characterizing the optical properties of each medium constituting the interface. k_0 is the incident laser wavenumber. t is the thickness of the thin layer sandwiched between environment and bulk sample [see Figure 1].

C. Optical modeling

Generally, we use dielectric function given by applying the Drude approximation to the electron gas and obtain [7]

$$\varepsilon = \varepsilon_\infty - \frac{\omega_p^2}{\omega^2} \frac{1}{1 + i\frac{\nu}{\omega}} \quad (6)$$

where

$$\omega_p^2 = \frac{n_e e^2}{m_e m^* \varepsilon_0}$$

Parameters are detailed in Table I. Description of water can be further improved by considering Refs. [11], and also Niemz (book)).

II. EXPERIMENTAL RESULTS

The presented experimental results only focus on LSFL structures. Indeed, it is believed that HSFL structures (far from laser wavelength) are not due to the excitation of Surface Plasmon Polariton. The experimental measurements used as a basis in this work are presented in Table II.

III. THEORETICAL RESULTS

A. Periodicity of SPP at single interface

Periodicity calculations have been performed for possible combinations between the following interfaces: water, excited water, Si, excited Si, and amorphous Si.

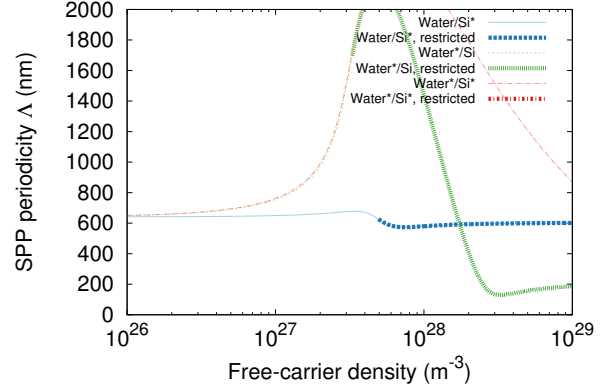


Figure 2: Periodicity of Surface Plasmon Polariton at the interface $Water^*/Si$

Periodicity of SPPs at $Water/a - Si^*$ and $a - Si^*/Si$ interfaces have been performed by Miyaji et al. [9] and will be used in the next sections.

However, calculations had not been performed for excited water interface with Si. This case is interesting and can lead to generation far subwavelength LSFL ripples perpendicular to the polarization as shown on Figure 2. Water can be excited by laser pulse and generate a high number of free-carriers ($N_e \sim 3 \times 10^{22} \text{ cm}^{-3}$). Indeed, the Kerr effect in water is ## times stronger than in air ($n_2^{water} =$) and the number of carriers in valence band is not limited in the liquid [11, 12].

B. Verification of the multilayer model

a. Checking the limit case $a \rightarrow 0$ In case of two identical interfaces, the SPP wave number is given by Eq. 3. Performing analytical resolution of dispersion relation (Eq. 4) over the multilayer model leads to the analytical solutions,

$$\beta_{a=0} \in \left\{ \pm k_0 \sqrt{\varepsilon_1}, \pm k_0 \sqrt{\frac{\varepsilon_2 \varepsilon_3}{\varepsilon_2 + \varepsilon_3}} \right\}$$

The first solution describes a propagation mode, and the second describe a SPP mode at a single interface made by media II and III. Thus, calculating the wave number β numerically can lead to the degeneracy that we well identified analytically. The degeneracy of the solution can lead to a discontinuous resolution using the numerical resolution technique.

b. Validation on previous works First, the model has been verified by reproducing the results of Prade et al. [6]. The results were reproduced with a very good accordance [see theoretical report on HSFL in Ti].

We also consider the case of experimental works proposed by Miyaji et al. [9]. Figure 3 presents the SPP periodicity as a function of the free-carrier density N_e . Periodicity is demonstrated for a single interface SPP living at the $water/a - Si^*$ interface and at

Material	Medium	N	Period	Orientation to $\vec{E}$
Si	air	100	555 – 585 nm	$\perp$
Si	water	100	80 – 130 nm	$\perp$
Ti	air	50	590 – 80 nm	$\perp$
Ti	water	1000	70 – 170 nm	$\perp$

Table II: Experimental periods of ripples on Si and Ti, in air and water.

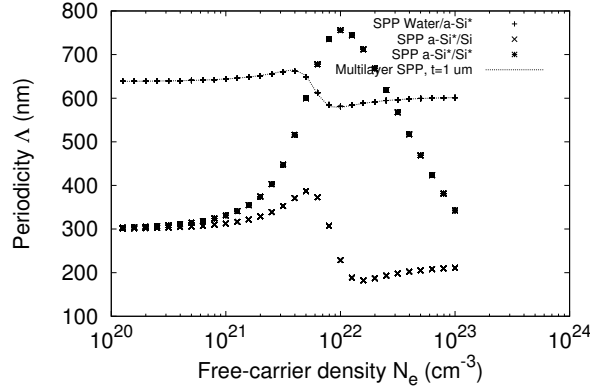
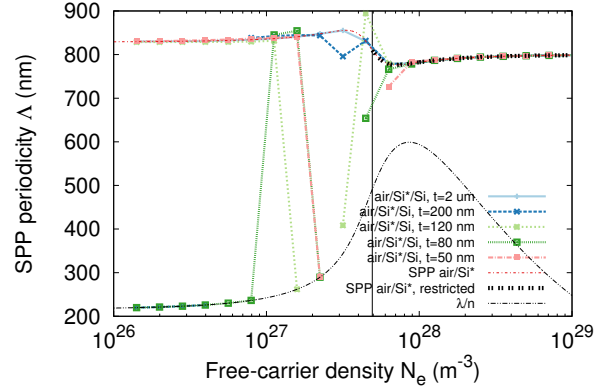


Figure 3: Periodicity of the SPP in the case of $Water - aSi^* - Si$ interface. Both single SPP model and multilayer SPP model are plotted. For this Figure, $\epsilon_{\infty}^{Si} = 13.5 + 0.0384i$ and $\epsilon_{\infty}^{a-Si} =$ were taken from Ref. [9].

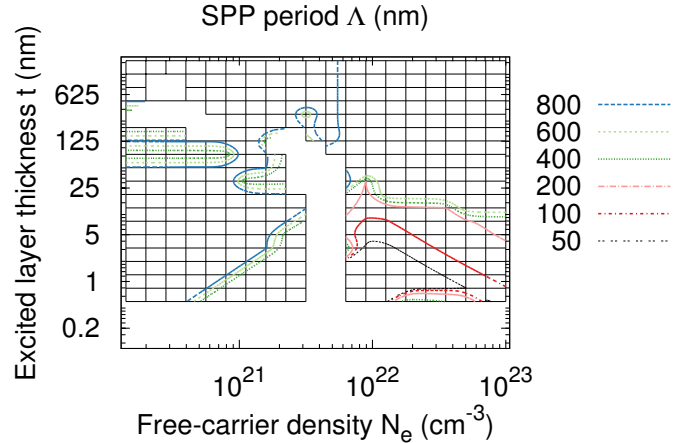
the $a - Si^*/Si$ interface. It is also calculated at the $a - Si^*/Si^*$ interface. The multilayer SPP results are also shown for the case of a large layer thickness ($t = 1 \mu m$ here). (i) Comparison with Ref. [9] reveals that invoking “standing-wave model” allowed to divide the results by a factor 2. However, the interference between SPP and laser is included in the dispersion relation of SPPs and thus should be avoided. (ii) The multilayer SPP results match well with the case of SPP living at the $water/a - Si^*$ interface. This allows us to validate the results obtained with multilayer model and to go deeper using different configurations. Only the $water/a - Si$ interface leads to SPP excitation since $\alpha^{-1}(a - Si) < t$ where t is the distance between interfaces. (iii) None of the previously obtained results well explain the experimental periodicity of Si irradiated in water.

C. The effect of coupled-SPPs in a thin layer

In the previous section, we reproduced the results of a SPPs living at a single interface $water/a - Si^*$ using a multilayer model, considering the full sandwich $water/a - Si^*/Si$, and setting the thickness of the $a - Si^*$ layer to a large value e.g. $1 \mu m$, which validates the approach. We apply now the multilayer model to simpler systems in order to understand the results obtained in a case a thin film. Then, we will apply the



(a)



(b)

Figure 4: (a) Periodicity at $air - Si^*$ interface. (b) Isolines representing the SPP modes as a function of the free-carrier density in $Air - Si^* - Si$ system. Si^* is a Si excited layer. The case of $air - Si^*$ has been developed in Ref. [13].

multilayer model to a more complex configuration.

1. The case of $air/Si^*/Si$ revisited

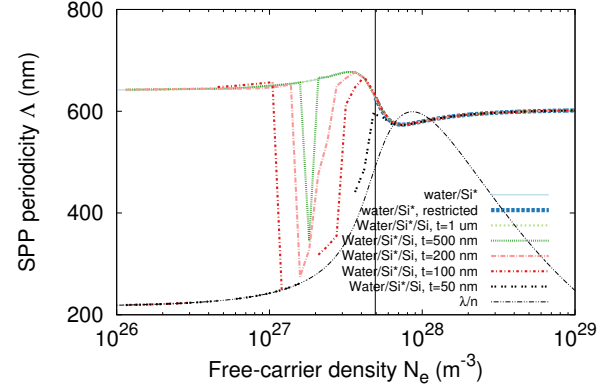
In this section, we analyze the simple case of $air - Si$ with a thin excited layer at the surface.

Figure 4 (a) presents the periodicity of the SPPs at interfaces with excited Si. The periodicities of SPP excited at single interface between air and excited Si is presented. We also propose a “restricted SPP” regime, taking into account densities for which SPP excitation is allowed considering generalized conditions valid in materials with losses (Eq. 2), compatible with recently published works of Norrman et al. [5] (see Figures therein). The proper conditions have not been published yet (LET’S DO IT! SEE REPORT ON T1 HSFL). It is demonstrated that the periodicity of SPP at single interface *air* – *Si** is well described by the multilayer model with a large layer thickness. The Figure also demonstrates that multilayer SPP model well reproduces results of single layer SPPs, which means that no SPP is excited at the second interface if $t = 2 \mu\text{m}$. As a verification, the case of large thickness is found equal to the single *air* – *Si** interface SPP periodicity. While $t = 200 \text{ nm}$, we observe that a periodicity of nearly 210 nm can be obtained at low density, and which corresponds to the wave propagation regime since periodicity is equal to $\lambda/\text{Re}\sqrt{\epsilon}$. This case is consistent with the experimental observation of Costache [14] in Si. However, this mode is not really of SPP since conditions to excite SPPs are a high density in the conduction band. This mode is found at sub-critical densities and would correspond to interferences between propagated waves (radiation remnants ?). At lower thicknesses, the multilayer SPP calculations provide an interesting possible transition between the SPP mode and the propagation mode. The transition occurs at the critical density defined by

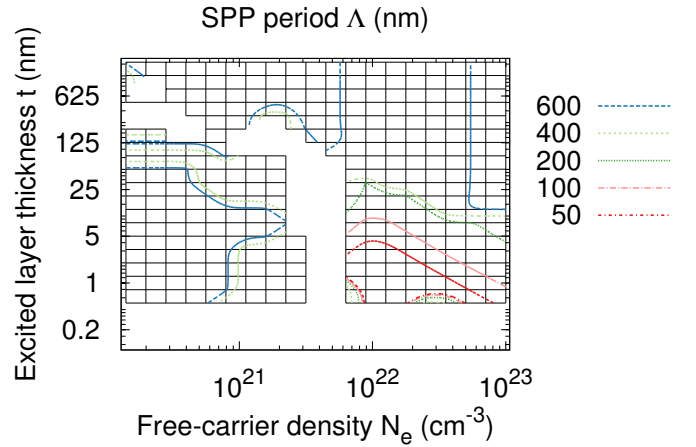
$$n_{cr} = \frac{m_e \epsilon_0 \text{Re}(\epsilon_\infty)}{e^2} (\omega^2 + \nu^2) \quad (7)$$

which has been represented by a vertical bar located at $n_{cr} = 4.93 \times 10^{27} \text{ m}^{-3}$.

Figure 4 (b) reveals the periodicity of SPP living in the *Si** layer in the *air* – *Si** – *Si* multilayer system. While reducing the layer thickness, the periodicity in the thin layer changes due to the interferences between the SPP living at the different interfaces. While reducing the layer thickness, the SPP periodicity decreases and reveals modes with a sub-100 nm periodicity. In the case of irradiation with a 100 fs laser, these modes have not been observed since the absorption depth of Si under ablation fluences is rather 100 nm [4, 13]. Considering that limit, the periodicities in Si are higher than 600 nm according to the results, which is consistent with the experimental observations [15, 16] at reduced number of pulses. At higher number of pulses, the cumulative effects could lead to a reduction of the skin depth which could explain the tiny ($\sim 200 \text{ nm}$) periodicities at very high number of pulses [14]. Moreover, one has to consider the excitation conditions which restrict the possible densities for excitation.



(a)



(b)

Figure 5: (a) Periodicity at *water* – *Si** interface. (b) Isolines representing the SPP modes as a function of the free-carrier density in *water* – *Si** – *Si* system. Vertical bar shows the value of critical density in Si medium.

2. The effect of interaction in water

We turn now to the study of configurations (I) *water* – *Si** – *Si* and (II) *water* – *water** – *Si*.

Figure 5 presents the case of the *Water* – *Si** – *Si* interface. At $1 \mu\text{m}$ thickness, periodicity calculated with multilayer and single SPP models are in good accordance. This means that the first interface behaves as a bulk under laser interaction and supports the SPP excitation. The second interface does not see the laser excitation. The interference between the SPPs is possible only if they are close enough. This threshold distance is related to the skin depth of the layer under laser excitation. We can observe a transition from SPP regime to propagation regime while free-carrier density is reducing. The criterion to excite SPP then remains important

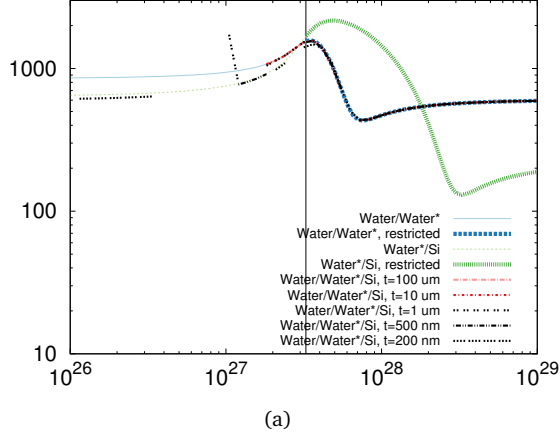


Figure 6: Isolines representing the SPP modes as a function of the free-carrier density in $Water - Water^* - Si$ system. Vertical bar shows the critical density in water medium ($n_{cr}^{H_2O} = 3.26 \times 10^{27} m^{-3}$).

in the case of a multilayer system and corresponds to a threshold given by the critical density (Eq. 7). Finally, this configuration still does not explain the experimental results, forcing us to check other possibilities.

The SPP periodicities are reduced in the case of an alternated nature of layers such as *dielectric – metal – dielectric* or *metal – dielectric – metal*. We turn now the study the possibility to excite water more than exciting the sample itself.

Figure 6 presents the simulation results performed considering the excitation of a layer of water between the ambient medium and the bulk Si.

Figure 6 (a) demonstrates the SPP periodicity at the $water - water^*$ interface, and $water^* - Si$ interface. The “restricted” curves show the periodicities for which SPP are allowed to be excited regarding condition given by Eq. (2). We first underline multilayer $water - water^* - Si$ system shows the same periodicity as SPP at single interface $water - water^*$. No low periodicity is observed for any layer thickness. However, the SPP at simple interface $water^*/Si$ reveals a periodicity of 105 nm near $N_e \sim 3 \times 10^{28} m^{-3}$. Vertical bar shows the critical density in water ($n_{cr}^{H_2O} = 3.26 \times 10^{27} m^{-3}$). However, the works of Noack and Vogel [11] revealed that such densities can be reached in our regime.

Considering the excitation of water revealed that sensitive reduction of the SPP periodicity can be reached and must be investigated more deeply. Moreover, looking forward alternating nature of dielectric layers drive us to consider an oxide layer, which has been demonstrated as important after irradiation in water.

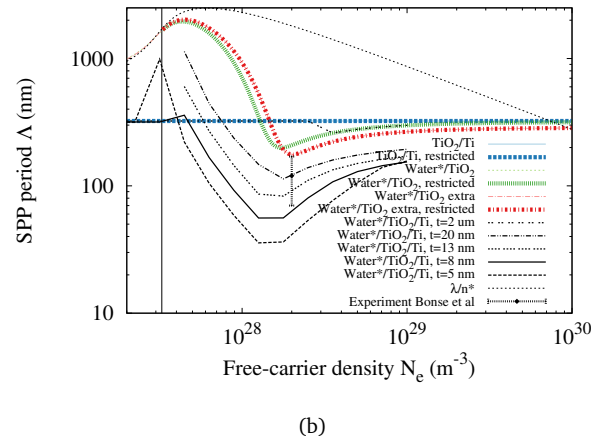
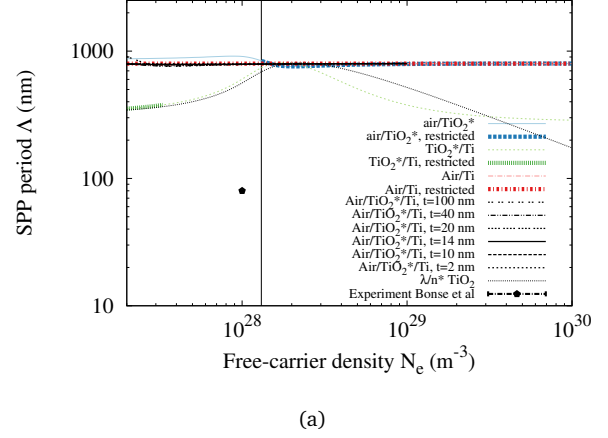


Figure 7: Periodicity in (a) the $air/TiO_2^*/Ti$ system, (b) $water^*/TiO_2/Ti$ systems

D. Interfaces involving oxide layer

Following the lead after the low periodicities provided by excitation of water, we turn now to the study of SPP periodicity considering the interfaces $water^*/oxide/bulk$ and compare with the periodicities obtained in air medium.

1. $air/TiO_2^*/Ti$ and $Water^*/TiO_2/Ti$ systems

Figure 7 (a) presents the possible EM wave mode periodicities in the multilayer system $air - TiO_2^* - Ti$. It is shown that no special subwavelength mode lives in the TiO_2^* layer.

Figure 7 (b) presents the possible EM mode periodicities in the multilayer system $water^* - TiO_2 - Ti$. It is demonstrated that SPP mode living at the interface $water^* - TiO_2$ can decrease to 200 nm, and down to 180 nm in the case of extraordinary axis of the crystalline orientation of TiO_2 . The periodicity of the SPP living in the TiO_2 layer is distributed around the TiO_2/Ti interface, since periodicity of SPP at this in-

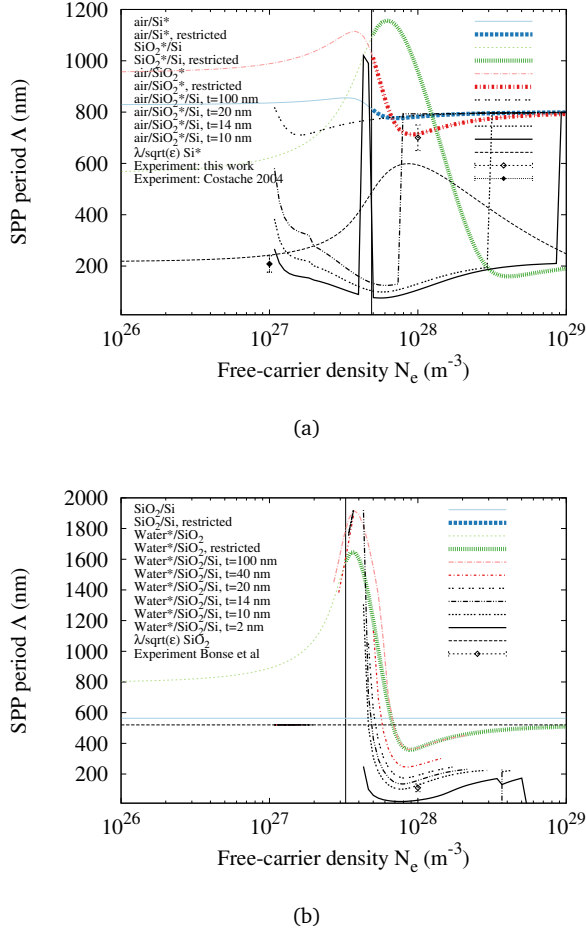


Figure 8: Periodicity in (a) the $air/SiO_2^*/Si$ system, (b) $water^*/SiO_2/Si$ systems

terface is 210 nm. By increasing density of water, the periodicity of SPP decreases in oxide layer, especially if the thickness of the oxide layer decreases to several nanometers. In that case, sub-100 nm periodicities become possible to reach. The experimental measurement reported in Table II has been added to the Figure 7 (b). These results demonstrate that considering the excitation of the surrounding medium e.g. water allows to explain the sub-200 nm periodicity in case of a presence of an oxide layer at the surface of Ti. A very good match is found for a oxide layer thickness of 20 nm, which is rather reasonable after several irradiations. Moreover, such periodicities can be reached according to Noack and Vogel [11].

2. $air/SiO_2^*/Si$ and $water^*/SiO_2/Si$ systems

Figure 8 (a) presents something that nobody will understand and must be wrong. CHECK.

Figure 8 (b) presents the periodicity of SPPs calculated for the configuration $Water^*/SiO_2/Si$. It is demonstrated that periodicity of SPPs can be found

lower than 100 nm in the case of a thin oxide layer and excited water. A good agreement is found with $t = 10$ nm for reachable densities $N_e = 8 \times 10^{27} m^{-3}$ in modification fluence regime. The periodicity as a function of free-carrier density present in water shows the same behavior than SPP at the $water^*/SiO_2$ layer, but with a lower periodicity, which suggest that presence of the Si surface below thin film is important and plays a role in the reduction of the periodicity.

As a conclusion of our investigations, we obtained two configurations matching with experimental results (on Si and Ti irradiated in water) by considering two key points: (i) excitation of the surrounding medium (water in this case), (ii) presence of an oxide layer, which has been demonstrated as being important by chemical analysis. The multilayer system with excited water and oxide thus well explains the observed periodicities and constitute the first understanding of such a substantial periodicity reduction in liquids.

E. Possible number of free-carriers in Si

Now we have calculated possible periodicities at the interfaces present in the system, and considered the possible interactions together, we calculate the densities that the laser can excite in either Si, in air and in water medium. In this section, we calculate the number of carriers excited in the conduction band. Calculations were performed for interaction in air and in water.

Free-carrier density N_e is calculated considering photonic transitions by solving Eq. (8)

$$\frac{\partial N_e}{\partial t} = \frac{\sigma_1 I}{\hbar \omega} + \frac{\sigma_2 I^2}{2 \hbar \omega} \quad (8)$$

where laser intensity at the surface is given by Eq. (9)

$$I(t)_{z=0} = (1 - R) \times \frac{F}{\tau} \times e^{-\frac{1}{2} \left(\frac{t}{\sigma_\tau} \right)^2} \quad (9)$$

where $\sigma_\tau = \frac{\tau}{2\sqrt{2 \ln 2}}$.

The calculations have been performed in three different cases: (i) by solving Eq. (8, 9) and taking R as constant; (b) by using the model of Ref. [13] but with Fresnel formulas to account for the surface reflectivity; (iii) to account for the contribution to the whole sample to the surface reflectivity. Figure 9 presents the maximum free-carrier density as a function of laser fluence. It is demonstrated that surface screening becomes important above $0.15 J/cm^2$ while $\tau = 30$ fs and $\lambda = 800$ nm. The possible number of free-carriers is thus known at the $air - Si^*$ interface. The density of free-carriers has also been calculated the water environment. We demonstrate that more free-carriers are excited in the case of water environment comparatively to irradiation in air. Indeed, reflectivity at the interfaces $air - Si$ is 32.9%. However, the reflectivity at $H_2O - Si$ interface is 22%. However, this model does not take into account the losses at the $air - water$ interface. Experimental

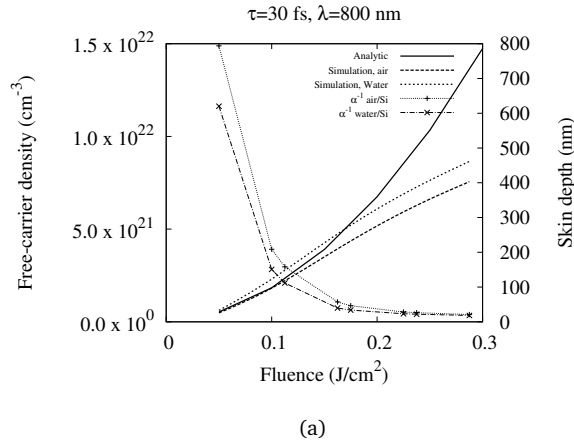


Figure 9: Maximum free-carrier density as a function of laser fluence, model and environment.

irradiations have been performed at 0.1 J/cm^2 , which demonstrate the possible number of free-carriers in Si surface in air environment is $2.0 \times 10^{21} \text{ cm}^{-3}$, and is $2.5 \times 10^{21} \text{ cm}^{-3}$ in water environment. There can be an uncertainty error bar in the fluence which drives the possible values to a larger. The skin layer at this fluence is 200 nm in air and 150 nm in water environments. It is also demonstrated that at this laser fluence, the analytic calculations are sufficient e.g. surface reflectivity is not modified during laser irradiation.

IV. CONCLUSIONS

A multilayer SPP model has been applied to different configurations. In each case, it has been demonstrated that the large thickness case leads to reproduce the single interface SPP, which validates the model for deeper investigations. Moreover, the case of Si excited in air environment has been revisited. It is revealed that the transition between propagative and SPP modes is well-described by the dispersion relation of possible modes in the layer. The dispersion relation has been calculated numerically and contains propagative mode in λ/n^* and SPP mode. The conditions of SPP mode excitation $\text{Re}(\varepsilon) = 0$ also apply in the case of a multilayer system and the required threshold density corresponds to the critical density.

Additionally, sub-200 nm periodicities shown by LIPSS formed at the surface of Si and Ti in water environment are explained by excitation of Surface Plasmon Polariton at the *water*/oxide* interface, considering excitation of the water by the laser. It has been demonstrated that the resulting periodicities explain for the first time the observed “tiny” LSFL structures at the interface of Ti and Si irradiated in water. Such conditions of irradiation allow a simple way to control the periodicity of the structure far from diffraction limit and are explained by the excitation of Surface Plasmon Polaritons. The role of the oxide layer and the excitation of the surrounding medium are responsible for the reduction of the SPP periodicity, by reduction of the periodicity through interference between SPP modes at the interfaces with the waveguiding modes in the oxide layer.

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- [1] Heinz Raether. *Surface plasmons on smooth and rough surfaces and on gratings*. Springer-Verlag, 1986.
 - [2] AV Zayats, II Smolyaninoy, and AA Maradudin. Nano-optics of surface plasmon polaritons. *Physics Reports*, 408:131–314, 2005.
 - [3] Stefan Alexander Maier. *Plasmonics, Fundamentals and Applications*. Springer, 2007.
 - [4] Thibault J.-Y. Derrien. *Nanostructuration de cellules photovoltaïques par impulsion laser femtoseconde. Etude des mécanismes de formation*. PhD thesis, Université de la Méditerranée - Aix Marseille II, Février 2012.
 - [5] Andreas Norrman, Tero Setälä, and Ari T. Fridberg. Exact surface-plasmon polariton solutions at a lossy interface. *Optics Letters*, 38:7, 2013.
 - [6] B. Prade, J.Y. Vinet, and A. Mysyrowicz. Guided optical waves in planar heterostructures with negative dielectric constant. *Physical Review B*, 44:24, 1991.
 - [7] M.S. Dresselhaus. *Solid State Physics. Optical properties of Solids*. MIT University, 2001.
 - [8] K. Sokolowski-Tinten and D. von der Linde. Generation of dense electron-hole plasmas in silicon. *Physical Review B*, 61:4, 2000.
 - [9] Godai Miyaji, Kenzo Miyazaki, Kaifeng Zhang, Takakazu Yoshifuji, and Junya Fujita. Mechanism of femtosecond-laser-induced periodic nanostructure formation on crystalline silicon surface immersed in water. *Optics Express*, 20:14, 2012.
 - [10] C. Sarpe, J. Köhler, T. Winkler, M. Wollenhaupt, and T. Baumert. Real time observation of transient electron density in water irradiated with tailored femtosecond laser pulses. *New Journal of Physics*, 14:075021, 2012.
 - [11] Joachim Noack and Alfred Vogel. Laser-induced plasma formation in water at nanosecond to femtosecond time scales - calculation of thresholds, absorption coefficients, and energy density. *IEEE Journal of Quantum Electronics*, 35:8, 1999.
 - [12] Alfred Vogel, Joachim Noack, Gereon Hüttmann, and Günther Paltauf. Femtosecond-laser-produced low-density plasmas in transparent biological media: A tool for the creation of chemical, thermal and thermomechanical effects below the optical breakdown threshold. *Proceedings of SPIE*, 4633:23–37, 2002.
 - [13] Thibault J.-Y. Derrien, Tatiana E. Itina, Rémi Torres, Thierry Sarnet, and Marc Sentis. Investigation of the possibility of surface plasmon polariton excitation under femtosecond laser irradiation of silicon in the regime of periodic surface structure formation. *Journal of Applied Physics*, Accepted, 2013.
 - [14] Florentina Costache. Sub damage threshold femtosecond laser ablation from crystalline *si*: surface nanos-

- tructures and phase transformation. *Applied Physics A*, 79:1429, 2004.
- [15] J. Bonse, A. Rosenfeld, and J. Krüger. On the role of surface plasmon polaritons in the formation of laser-induced periodic surface structures upon irradiation of silicon by femtosecond-laser pulses. *Journal of Applied Physics*, 106:104910, 2009.
- [16] Jörn Bonse and Jörg Krüger. Pulse number dependence of laser-induced periodic surface structures for femtosecond laser irradiation of silicon. *Journal of Applied Physics*, 108:034903, 2010.